Wi-Fi Security Intelligence & Monitoring Platform

Complete Platform Features & Architecture Guide (Plain-English Edition)

Created: October 20, 2026 | 100% Passive & Defensive Scope | Platform: Windows 10/11

Executive Overview: What is this project?
Wi-Fi networks operate invisibly through the air, making it easy for attackers to set up deceptive traps, eavesdrop on unencrypted traffic, or hijack active internet connections. This software acts as an automated **digital security guard** for your wireless environment. It continuously listens to the radio waves, checks for dangerous red flags, verifies that trusted networks haven't been cloned, and alerts you immediately if something is wrong—all without transmitting noisy packets or violating privacy.

Why this guide exists: Even though the code under the hood performs complex mathematical modeling, IEEE 802.11 frame analysis, and cryptographic scoring, understanding it shouldn't require a master's degree in cybersecurity. This guide explains every single functionality using plain, simple English, real-world analogies, and clear step-by-step reasoning.

Part 1: The Top 5 Core Cybersecurity Superpowers

Feature 1: Evil Twin & Rogue Access Point Detection [NEW CRITICAL DEFENSE]

■ What Is It?
An attacker sets up a fake, cloned Wi-Fi router broadcasting the exact same name (SSID) as your legitimate home, college, or office network (e.g., *'VIT-Campus'* or *'Home_WiFi'*). When devices automatically connect to this clone, the attacker intercepts their passwords, browsing traffic, and credentials.

■ Why Does It Matter?
Smartphones and laptops are designed to remember network names. If an attacker sits in a parking lot with a high-powered fake router named after your campus or home, your laptop may seamlessly hand over its traffic to the hacker without your knowledge.

■■ How Does It Work? (Simple Explanation)
Analogy: It's like a stranger wearing a nametag with your best friend's name, but having a completely different face and voice.

1. **Mark as Trusted:** You tell the platform your real router once. The tool memorizes its unique hardware MAC address (BSSID), manufacturer (e.g., TP-Link), security encryption (WPA2), and channel.

2. **Forensic Evidence Scoring:** Every time the scanner runs, if it spots a network with a trusted name but unfamiliar attributes, it calculates an *Evil Twin Evidence Score* using 5 explainable factors:

- **Same SSID, Unknown BSSID (+40 points):** An unrecognized hardware radio is impersonating your network name.
- **Security Downgraded (+30 points):** The real network requires a password (WPA2), but this twin has no password (Open) to lure victims in.
- **Different Hardware Manufacturer (+15 points):** Real router is made by Cisco, but this one is an unbranded chip or Raspberry Pi.

• **Different Channel (+5 points):** The imposter transmits on an unexpected radio channel.

• **Fake / Randomized MAC (+10 points):** The MAC address has the IEEE 'locally administered' bit turned on, indicating software spoofing.

3. **Instant Response:** If the score reaches 40+, the tool triggers an urgent **CRITICAL Alert** to warn you before your laptop connects!

■ ***Where To Find It In The App:*** *Open* ***Baseline & Trust*** *page → Click* ■ ***Mark Selected as Trusted****. When an impostor appears, red banners light up on the Dashboard, Events log, and as a Windows desktop notification.*

Feature 2: Real-Time Multi-Channel Alerts & Anti-Spam Engine [NEW SMART NOTIFIER]

■ What Is It?

An automated notification system that alerts you the second a threat appears, even if the app is minimized, running in the background, or you are away from your desk.

■ Why Does It Matter?

A security monitor that quietly writes warnings into a closed log file is useless if an attack occurs while you are browsing the web. You need to be alerted immediately so you can disconnect before data is compromised.

■■ How Does It Work? (Simple Explanation)

Multiple Delivery Channels:

• **Windows Toast Notifications:** Native popup banners in the corner of your screen. Built using Windows native APIs—works 100% offline without needing internet.

• **Telegram Bot & Email:** Sends instant messages to your smartphone so you stay protected even while away from your computer.

Smart Anti-Spam Controls (Keeps alerts helpful, not annoying):

• **Deduplication:** The same warning will never be sent twice in a row.

• **30-Minute Cooldown:** If an offending rogue AP stays on, you get warned once, then the system stays quiet for 30 minutes instead of chiming constantly.

• **Severity Threshold:** You choose your alert level (e.g., only High & Critical threats, or all alerts).

• **One-Click Acknowledge:** Clicking 'Acknowledge' in the alerts tab marks the event as seen and permanently stops repeated notifications.

■ ***Where To Find It In The App:*** *Configure thresholds and bot keys in* ***Settings*** *→* ***Alerts & Notifications****. View live alarms under* ***Events & Alerts*** *→* ***Real-Time Alerts****.*

Feature 3: Weak-Configuration & Default Password Auditor [NEW VULNERABILITY AUDITOR]

■ What Is It?

An automated vulnerability scanner that inspects every Wi-Fi network within radio range and pinpoints dangerous configuration weaknesses—even if no active attack is currently occurring.

■ Why Does It Matter?

Many home and small-business routers are left with outdated encryption standards or factory default names. Hackers actively drive around (wardriving) looking for these easy targets because they can be broken into in minutes.

■■ How Does It Work? (Simple Explanation)

Analogy: Like a home security inspector walking around checking if doors are unlocked or windows are left open.

The engine tests each network against 6 automated security rules:

• **Open Networks (+85 points, CRITICAL):** Zero encryption or password. Anyone sitting nearby can eavesdrop on your unencrypted traffic.

• **WEP Encryption (+75 points, HIGH):** Obsolete 1999 encryption standard that can be cracked in under 60 seconds with free software.

• **Legacy WPA (+35 points, MEDIUM):** Outdated security susceptible to dictionary password attacks.

• **TKIP Cipher (+20 points, MEDIUM):** Deprecated encryption algorithm that slows down modern Wi-Fi and has known mathematical flaws.

• **Hidden SSIDs (+15 points, LOW):** Hiding a network name provides zero real security; devices constantly broadcast search probes in public to find it.

• **Default Router Names (+15 points, LOW):** Factory names like *'TP-Link_4F2A'*, *'JioFiber-9912'*, or *'NETGEAR62'*. These tell hackers the exact hardware model and suggest the default router admin password was never changed!

■ ***Where To Find It In The App:*** *Double click any network in* ***Live Scanner*** *→ Open the* ***Weak Config*** *tab to see full findings and point breakdowns.*

Feature 4: Channel Recommendation Engine (2.4 GHz & 5 GHz) [NEW SPEED & STABILITY OPTIMIZER]

■ What Is It?

An intelligent radio frequency analyzer that tests Wi-Fi airwave congestion and tells you the exact channel number your router should switch to for maximum speed and zero lag.

■ Why Does It Matter?

Wi-Fi signals travel through radio frequencies like cars on highway lanes. On the 2.4 GHz band, channels heavily overlap. If your neighbor is on channel 6 and you are on channel 5, your routers transmit over each other, causing gaming lag, buffering, and slow downloads.

■■ How Does It Work? (Simple Explanation)

A smart traffic GPS for your Wi-Fi:

1. **The 2.4 GHz Overlap Problem:** Only channels **1, 6, and 11** do not physically interfere with each other. All other channels (2, 3, 4, 5, 7, etc.) cause adjacent collisions. The engine calculates an exact **Congestion Score** for 1, 6, and 11 by adding the signal strength of all routers sharing the channel, plus **70%** of the signals from overlapping neighboring channels:
   • *Formula:* Congestion = Sum(Co-Channel Signal) + 0.7 × Sum(Adjacent Signal)

2. **The 5 GHz Band:** 5 GHz channels do not overlap! The tool counts networks on each channel (36, 40, 44, 48, etc.) and picks the cleanest, emptiest lane.

3. **Clear Recommendation:** Instead of confusing graphs, it gives you a simple verdict: *'Channel 6 has 9 networks (congestion score 480). Channel 11 has 1 network (score 42). Recommended: Switch router to Channel 11.'*

■ ***Where To Find It In The App:*** *Navigate to* ***Analytics & Signals*** *→ Look at the prominent card at the top titled* ***Channel Recommendation Engine*** *with live 2.4 GHz and 5 GHz badges.*

Feature 5: Active Connection Integrity (ARP & DNS Tamper Guard) [NEW CONNECTION INTEGRITY WATCHDOG]

■ What Is It?

A dedicated watchdog that actively protects the specific Wi-Fi network your computer is *currently connected to*, guarding against Man-In-The-Middle (MITM) and DNS diversion attacks.

■ Why Does It Matter?

Scanning surrounding routers is helpful, but what if the network you are currently working on gets hacked? If an attacker on the same café or office Wi-Fi poisons your computer's routing table (ARP spoofing) or tampers with your DNS, they can silently steer your browser to fake phishing sites.

■■ How Does It Work? (Simple Explanation)

Two continuous watchdog checks running quietly in the background:

1. **ARP Spoofing Detection:** When you join a Wi-Fi network, your computer communicates with your router (the Default Gateway). The monitor memorizes the router's physical MAC address. If a hacker attempts ARP poisoning to trick your laptop into sending traffic through their laptop instead, the gateway MAC address suddenly changes. The monitor immediately flags an urgent **CRITICAL Man-In-The-Middle alert**.

2. **DNS Hijacking Detection:** When you type 'google.com', your computer asks a DNS server for the IP address. The monitor periodically resolves canary domains (Google, Cloudflare, Wikipedia) and compares the responses against tamper-proof **Cloudflare DNS-over-HTTPS (DoH)**. If the IP addresses differ or point to a local private decoy address, the monitor warns you that your internet is being hijacked!

■ ***Where To Find It In The App:*** *Visible right on the **Dashboard** in the **Connection Health** card. You can also click **Check Connection Health** for an instant on-demand audit.*

Part 2: The Core Defensive & Monitoring Foundation

Feature 6: 100% Passive Wi-Fi Discovery Engine [CORE ENGINE]

■ **What Is It?**

The underlying scanner that quietly discovers all nearby Wi-Fi networks (names, MAC addresses, signal strength, channel frequencies, and security protocols) using native Windows wireless services.

■ **Why Does It Matter?**

Many offensive hacking tools transmit illegal deauthentication frames or intrusive probe requests. This platform is 100% passive: it operates like an FM radio tuner, silently listening to public beacon signals without transmitting a single packet or alerting anyone.

■■ **How Does It Work? (Simple Explanation)**

1. Invokes the Windows WLAN interface query (`netsh wlan show networks mode=bssid`) every few seconds.
2. Runs custom high-speed regular expressions to extract SSID, BSSID (MAC), Signal %, Authentication, Cipher, and Channel.
3. Works completely offline without needing special hardware, Wi-Fi dongles, or custom drivers.

■ ***Where To Find It In The App:*** *Runs automatically. View live nearby radios anytime in the **Live Scanner** tab.*

Feature 7: Explainable Multi-Factor Risk Engine (0-100 Score) [CORE SCORING]

■ **What Is It?**

A mathematical scoring model that grades every Wi-Fi network's safety on a clear 0 to 100 scale (0 = Completely Safe, 100 = Extremely Dangerous).

■ **Why Does It Matter?**

Standard operating systems only show a generic lock icon. They never explain whether 'WPA2-Personal' is safe, or how much danger you are in when connecting to an open hotel network.

■■ **How Does It Work? (Simple Explanation)**

Calculates an explainable score by evaluating transparent security factors:

- **Security Protocol:** Open network adds +85; WEP adds +75; WPA adds +35; WPA2 adds +10; WPA3 adds 0.
- **Cipher Quality:** Weak cipher (TKIP) adds +20; strong cipher (CCMP/AES) adds 0.
- **Configuration Flaws:** Hidden network (+15); default router name (+15); security downgrade (+30).
- **Four Color-Coded Tiers: LOW** (0–29), **MEDIUM** (30–59), **HIGH** (60–79), and **CRITICAL** (80–100).

■ ***Where To Find It In The App:*** *Color-coded badges (Green, Yellow, Orange, Red) appear in **Live Scanner**, **Security Monitor**, and executive reports.*

Feature 8: Three-State Lifecycle Tracking & Debouncing [SMART STATE MANAGEMENT]

■ What Is It?
A state tracking system that remembers whether a network is genuinely present, temporarily flickering due to radio interference, or completely gone.

■ Why Does It Matter?
Wi-Fi signals fluctuate constantly when people walk by or doors close. Cheap Wi-Fi scanners send endless spam alerts because a network flickered out for a single second. Our system eliminates false alarms.

■■ How Does It Work? (Simple Explanation)
A 3-strike verification rule:

- **ACTIVE:** The network is clearly visible in the current scan.
- **TEMPORARILY_MISSING:** If a network disappears for 1 scan, the system does not panic. It starts a countdown and holds off alerting.
- **NOT_DETECTED:** Only when an access point is missing for **3 consecutive scans** is it officially marked offline. This simple debouncing rule eliminates over 99% of false alarms!

■ ***Where To Find It In The App:*** *Visible in the status column of the* ***Security Monitor*** *and logged inside the* ***History*** *view.*

Feature 9: Environment Baseline & Deviation Detection [ENVIRONMENT SNAPSHOT]

■ What Is It?
Allows you to take a saved 'Snapshot' of all normal, expected Wi-Fi networks in your home or office, and automatically alerts you when unexpected new radios appear.

■ Why Does It Matter?
If you have 3 authorized routers in your office, and one morning a 4th unexpected router starts broadcasting from a nearby vehicle or hallway, you need to know immediately.

■■ How Does It Work? (Simple Explanation)
1. Click **Create Baseline** while in your normal environment. The app snapshots every known BSSID, SSID, and channel into the database.
2. On all subsequent scans, every detected network is checked against this baseline photo.
3. Any unknown hardware or unexpected configuration shift generates a `BASELINE_DEVIATION` event.

■ ***Where To Find It In The App:*** *Access via the* ***Baseline & Trust*** *window in the sidebar menu.*

Feature 10: 8-Pattern Passive Anomaly Detection Engine [BEHAVIORAL HEURISTICS]

■ What Is It?
A pattern-matching engine that watches how wireless signals change over time to catch stealthy hacker behaviors.

■ Why Does It Matter?
Attackers frequently attempt subtle maneuvers like jamming a channel to force clients to switch, or dropping encryption temporarily to capture cleartext data.

■■ How Does It Work? (Simple Explanation)
The engine watches for 8 specific red flags:

1. **Security Downgrade:** Known network suddenly switches from WPA2 to Open.
2. **Channel Shift:** Router abruptly hops to a completely different radio channel.
3. **New BSSID on Known SSID:** A brand new hardware radio starts claiming a known network name.
4. **Signal Surge:** An AP's signal jumps from 20% to 95% in seconds (indicating a rogue device was brought close).
5. **Signal Drop:** Sudden signal loss indicating jamming or RF interference.

6. **High Density Alert:** Too many access points crowding a single frequency.
7. **Baseline Deviation:** An unknown network appearing in an established environment.
8. **Evil Twin Impersonation:** Impersonation score exceeding threshold.

■ ***Where To Find It In The App:*** *Review all flagged anomalies under* ***Events & Alerts*** → ***Anomaly Log****.*

Feature 11: Offline Hardware Vendor (OUI) Resolver [HARDWARE INTELLIGENCE]

■ What Is It?

Identifies the real physical manufacturer of any Wi-Fi router (e.g., Apple, Cisco, TP-Link, Intel, Raspberry Pi) without needing an internet connection.

■ Why Does It Matter?

A router may have a friendly name like 'Corporate_Secure', but its hardware MAC address reveals the real maker. If 'Corporate_Secure' is supposedly an enterprise Cisco router, but the tool shows the maker is 'Espressif' (a cheap $3 microchip) or 'Unknown / Fake', you know it's an imposter!

■■ How Does It Work? (Simple Explanation)

1. The first 3 bytes of any network MAC address are an IEEE-assigned Organizationally Unique Identifier (OUI).
2. The platform carries an embedded offline lookup database containing thousands of registered manufacturers.
3. Queries resolve instantly in memory in less than 1 millisecond, with zero internet access required.

■ ***Where To Find It In The App:*** *Shown in the* ***Vendor*** *column in Live Scanner, Security Monitor, and Network Details.*

Feature 12: Offline Cyber Simulation & Demo Mode [TRAINING & TESTING]

■ What Is It?

A safe virtual testing environment that lets you simulate cyber attacks (Evil Twins, rogue APs, security downgrades) without altering real-world routers.

■ Why Does It Matter?

Testing a security application in the real world can be difficult because you don't always have a spare rogue router or hacking equipment. Demo mode lets students, professors, and testers verify that all alerts and triggers work perfectly without real risk.

■■ How Does It Work? (Simple Explanation)

1. When enabled, the application generates realistic simulated Wi-Fi scans using virtual radio models.
2. Injects intentional vulnerabilities (e.g., an evil twin impersonating your network, or an AP suddenly dropping encryption).
3. Lets you verify that the risk engine, visual charts, and desktop notifications react properly without any physical radio interference.

■ ***Where To Find It In The App:*** *Toggle anytime in* ***Settings*** → ***Simulation Mode****.*

Part 3: Reports, Data Management & Modern Interface

Feature 13: Executive Security Reports (HTML, CSV, JSON) [EXECUTIVE REPORTING]

■ What Is It?

Generates comprehensive, professional security audit reports in HTML, CSV spreadsheet, and JSON data formats with a single click.

■ Why Does It Matter?

IT managers, professors, and cybersecurity auditors need tangible proof of security posture. Manually compiling scan logs takes hours; one-click reports can be shared or printed instantly.

■■ How Does It Work? (Simple Explanation)

1. Aggregates the latest scan data, historical anomaly logs, risk scores, and connection integrity status.
2. Formats the data into a beautifully styled, self-contained HTML report featuring responsive styling and charts.
3. Also exports raw data to CSV (for Excel analysis) and JSON (for integration with SIEMs or other tools).

■ ***Where To Find It In The App:*** *Click* ***Reports & Export*** *in the sidebar → Choose HTML, CSV, or JSON.*

Feature 14: Automated SQLite Migrations & Backup Controls [ENTERPRISE STORAGE]

■ What Is It?

An enterprise-grade local SQLite database that stores your history, baselines, and alerts, featuring automatic schema upgrades and safe backup/restore tools.

■ Why Does It Matter?

When software updates add new features (like our new `trusted_networks` or `alerts` tables), old user databases can break. Automated migrations prevent data corruption and ensure painless upgrades.

■■ How Does It Work? (Simple Explanation)

1. Tracks database versions using a built-in schema migration table (`schema_version`).
2. On startup, automatically checks if newer tables are needed and applies migrations without destroying user history.
3. Includes one-click timestamped backups and database restore tools directly in the UI.

■ ***Where To Find It In The App:*** *Navigate to* ***Database & Maintenance*** *in the sidebar to create backups or restore.*

Feature 15: Modern Multi-View Sidebar Desktop GUI [INTUITIVE USER INTERFACE]

■ What Is It?

A clean, dark/light-themed graphical desktop interface built with Python's native Tkinter, featuring a persistent navigation sidebar and responsive widgets.

■ Why Does It Matter?

Command-line tools are intimidating for non-experts. A modern, organized visual dashboard makes enterprise-grade wireless intelligence accessible to everyone.

■■ How Does It Work? (Simple Explanation)

Organized into 11 purpose-built screens:

- **Dashboard:** High-level status cards, network counts, connection health widget.
- **Live Scanner:** Real-time table of all nearby Wi-Fi radios.
- **Security Monitor:** Risk ratings, protocol breakdowns, and weak configuration tags.
- **History:** Historical log of all network sightings over time.
- **Events & Alerts:** Anomaly events and real-time alert notifications.
- **Analytics:** Signal density charts and channel recommendations.
- **Baseline:** Environment snapshots and trusted networks management.
- **Reports, Database, Settings, Diagnostics:** Full operational control.

■ ***Where To Find It In The App:*** *The main window when you launch `python main.py`.*

Part 4: Frequently Asked Questions & Quick Reference

■ Can this tool hack into or disconnect my neighbor's Wi-Fi?

No. This is a 100% defensive and passive security application. It never transmits deauthentication frames, never injects packets, and never attempts to crack passwords. It strictly acts as a protective shield and audit tool.

■ Does the tool require an internet connection to work?

No. The core scanner, offline vendor OUI database, risk scoring engine, baseline system, SQLite database, and Windows toast alerts operate 100% offline. Internet is only used if you optionally enable Telegram notifications or the DoH canary check.

■ What should I do if the tool gives me a CRITICAL Evil Twin alert?

Immediately disconnect from that Wi-Fi network! Check your router hardware. If someone nearby is broadcasting a fake copy of your Wi-Fi name, do not enter passwords or sensitive credentials until the rogue access point disappears.

■ Why does the tool flag my home router for having a 'Default Router Name'?

If your Wi-Fi is named *'TP-Link_XXXX'* or *'JioFiber-XXXX'*, an attacker knows exactly what brand you own and can guess default administrative passwords. Renaming your network to something unique (like *'Skyline_Guest'*) fixes this issue immediately.

Complete Feature Summary Matrix:

#	Feature Name	Primary Purpose	Key Security Benefit
1	Evil Twin Detection	Spots cloned/counterfeit Wi-Fi	Prevents credential theft & phishing
2	Real-Time Smart Alerts	Instant notifications via Toast/Telegram	Notifies immediately with zero spam
3	Weak Config Checks	Audits open Wi-Fi, WEP, default names	Finds vulnerable routers before attackers
4	Channel Recommendation	Calculates signal-weighted congestion	Boosts Wi-Fi speed and cuts latency
5	Connection Health Guard	Monitors gateway MAC & DNS via DoH	Protects active connected session
6	Passive Wi-Fi Discovery	Listens to native Windows 802.11 beacons	100% legal, non-intrusive, zero noise
7	0-100 Risk Engine	Calculates explainable safety scores	Clear risk ratings for non-technical users
8	Lifecycle & Debouncing	Active / Temp Missing / Gone states	Eliminates false alarms from RF flickers
9	Environment Baseline	Saves reference snapshot of your space	Flags unauthorized newly appearing routers
10	8 Anomaly Patterns	Detects channel hops, signal jumps	Catches subtle, stealthy attacker activity
11	Hardware OUI Resolver	Translates MAC address to manufacturer	Identifies spoofed or unbranded hardware
12	Offline Demo Simulator	Injects mock attacks safely	Ideal for college demos and presentations
13	Executive Reports	Exports clean HTML, CSV, and JSON	Ready to share with managers & auditors
14	Database & Migrations	Automated SQLite schema management	Zero data loss across platform updates
15	Sidebar Desktop GUI	Modern 11-panel Tkinter interface	Intuitive navigation with zero command-line